ALGEBRA II · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

6 question sets • 180 questions

Page numbers match the Question Collection.
Every page shows its question ID (for example S40-F2-Q3).

CONTENTS

Set 1	**HSN-CN.A.1**	Write complex numbers in the form a + bi	3
Set 2	**HSN-CN.A.2**	Add and subtract complex numbers	40
Set 3	**HSN-CN.A.2**	Multiply complex numbers and simplify powers of i	77
Set 4	**HSN-CN.C.7**	Solve quadratic equations with complex solutions	114
Set 5	**HSN-CN.C.8**	Factor polynomials over the complex numbers	151
Set 6	**HSN-CN.C.9**	Use the Fundamental Theorem of Algebra	188

Write complex numbers in the form a + bi

MAIN	Algebra II main question	**HSN-CN.A.1**
BACKWARD 1	Simplify square roots of positive numbers	**HSN-RN.A.2**
BACKWARD 2	No real number squares to a negative number	**8.EE.A.2**
BACKWARD 3	Divide each term of a sum by a number	**6.EE.A.3**
FORWARD 1	Use equality of complex numbers to find unknown real values	**HSN-CN.A.1**
FORWARD 2	Represent complex numbers as points in the complex plane	**HSN-CN.B.4**

SET 1 • HSN-CN.A.1

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write √(−49) in the form bi, where b is a real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

5

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

7i

GRADING NOTE Form required: bi. √(−49) = √49 · √(−1) = 7i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

Write √(−18) in the form bi with b in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

6

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

3i√2 (that is, 3√2 i)

GRADING NOTE Form required: simplest radical form. √(−18) = √9 · √2 · i = 3√2 i. Not accepted: √18 i (not simplified).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

Write $6 - \sqrt{(-20)}$ in the form $a + bi$. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

7

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

$6 - 2\sqrt{5}\,i$; real part 6, imaginary part $-2\sqrt{5}$

GRADING NOTE Form required: $a + bi$ with both parts named. $\sqrt{(-20)} = 2\sqrt{5}\,i$. The imaginary part is the real number $-2\sqrt{5}$, not $-2\sqrt{5}\,i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

Write (−4 + √(−75)) / 2 in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

8

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

−2 + (5√3/2)i; real part −2, imaginary part 5√3/2

GRADING NOTE Form required: a + bi with both parts named. √(−75) = 5√3 i; divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

Write (10 + √(−64)) / 4 in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

9

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

5/2 + 2i; real part 5/2, imaginary part 2

GRADING NOTE Form required: a + bi with both parts named (5/2 or 2.5 accepted). √(−64) = 8i; 10/4 = 5/2 and 8/4 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 1

Simplify square roots of positive numbers • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify √50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

11

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5√2

GRADING NOTE Form required: simplest radical form. 50 = 25 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

Simplify √72.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

12

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6√2

GRADING NOTE Form required: simplest radical form. 72 = 36 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q2**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

Which is √12 in simplest radical form?

A. 2√3

B. 4√3

C. 3√2

D. 6√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

13

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2√3

GRADING NOTE $12 = 4 \cdot 3$, so $\sqrt{12} = 2\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

14

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 7$. A square root of a sum is not the sum of the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

Simplify √28.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

15

QUESTION ID

S1-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2√7

GRADING NOTE Form required: simplest radical form. 28 = 4 · 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-B1-Q5**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 2

No real number squares to a negative number • Grade 8 • Expressions & Equations • 8.EE.A.2

Find all real numbers x with $x^2 = 81$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

17

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 9 or x = −9

GRADING NOTE Any order. Both 9^2 and $(-9)^2$ equal 81.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

True or false?

$(-3)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

18

QUESTION ID

S1-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-3)^2 = (-3)(-3) = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Is there a real number x with $x^2 = -4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

19

QUESTION ID

S1-B2-Q3

TYPE

Short answer

CORRECT ANSWER

No: the square of any real number is 0 or positive.

GRADING NOTE Any equivalent reason (a positive times a positive and a negative times a negative are both positive).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

Which number has a square equal to 36?

A. -6

B. 18

C. $\sqrt{-36}$

D. -18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

20

QUESTION ID

S1-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. −6

GRADING NOTE $(-6)^2 = 36$. $\sqrt{-36}$ is not a real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(-7)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

21

QUESTION ID

S1-B2-Q5

TYPE

Short answer

CORRECT ANSWER

49

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 3

Divide each term of a sum by a number • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write (8 + 6√2) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

23

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 3√2

GRADING NOTE Any equivalent form. Divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (−6 + 4√5) / 2 as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

24

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−3 + 2√5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

True or false?

$(10 + \sqrt{7}) / 5 = 2 + \sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

25

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both terms must be divided: $(10 + \sqrt{7})/5 = 2 + \sqrt{7}/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

(12 − 9√3) / 3 equals

A. 4 − 9√3

B. 4 − 3√1

C. 12 − 3√3

D. 4 − 3√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

26

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4 − 3√3

GRADING NOTE 12/3 = 4 and 9√3/3 = 3√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

Write (−4 + 2√3) / (−2) as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

27

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2 − √3

GRADING NOTE Any equivalent form. −4/(−2) = 2; 2√3/(−2) = −√3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 1

Use equality of complex numbers to find unknown real values • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

x and y are real numbers and (x − 3) + 2yi = 5 − 8i.
Find x and y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

29

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 8, y = −4

GRADING NOTE Form required: both values. Equal complex numbers have equal real parts (x − 3 = 5) and equal imaginary parts (2y = −8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

a and b are real numbers and 3a + (b + 1)i = −12 + 4i. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

30

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

a = −4, b = 3

GRADING NOTE Form required: both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

True or false?

If x and y are real and x + yi = 7, then y = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

31

QUESTION ID

S1-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 7 = 7 + 0i, so the imaginary parts give y = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

p and q are real and (p + q) + (p − q)i = 10 + 2i. What is p?

A. 5

B. 6

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

32

QUESTION ID

S1-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE p + q = 10 and p − q = 2 give p = 6 and q = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

m is a real number. For what value of m is $(m^2 - 4) + (m + 3)i$ a real number? What real number is it then?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

33

QUESTION ID

S1-F1-Q5

TYPE

Short answer

CORRECT ANSWER

m = −3; the number is 5

GRADING NOTE Form required: both. The imaginary part m + 3 must be 0, so m = −3 and $m^2 - 4 = 9 - 4 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 2

Represent complex numbers as points in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

In the complex plane the horizontal axis is real and the vertical axis is imaginary. Give the point for −3 + 5i.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

35

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, 5)

GRADING NOTE Form required: an ordered pair (real part, imaginary part).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

Which complex number is represented by the point (4, −2) in the complex plane?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

36

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 − 2i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

Find the distance from 0 to the point 6 + 8i in the complex plane.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

37

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $\sqrt{(6^2 + 8^2)} = 10$ (the modulus $|6 + 8i|$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number lies on the horizontal axis of the complex plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

38

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real number a is a + 0i, the point (a, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

Which number lies on the vertical axis of the complex plane?

A. $-7i$

B. -7

C. $7 + i$

D. $1 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

39

QUESTION ID

S1-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $-7i$

GRADING NOTE $-7i = 0 - 7i$, the point $(0, -7)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

Add and subtract complex numbers

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Add and subtract integers and decimals	**7.NS.A.1**
BACKWARD 3	Write square roots of negative numbers with i	**HSN-CN.A.1**
FORWARD 1	Find real parameters that make a sum real or imaginary	**HSN-CN.A.2**
FORWARD 2	Add complex numbers as vectors in the complex plane	**HSN-CN.B.5**

SET 2 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Simplify (3 + 5i) + (−7 + 2i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

42

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

−4 + 7i

GRADING NOTE Form required: a + bi. 3 − 7 = −4 and 5i + 2i = 7i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

Simplify (6 − 4i) − (−2 + 9i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

43

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

8 − 13i

GRADING NOTE Form required: a + bi. Subtract both parts: 6 − (−2) = 8 and −4i − 9i = −13i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

Simplify (−1 + √(−16)) + (5 − √(−9)). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

44

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

4 + i

GRADING NOTE Form required: a + bi. (−1 + 4i) + (5 − 3i) = 4 + i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

$z_1 = 2.5 - i$, $z_2 = -1.5 + 3i$ and $z_3 = 4i$. Find $z_1 - z_2 + z_3$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

45

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

4 (that is, 4 + 0i)

GRADING NOTE Form required: a + bi or the real number 4. Real: 2.5 + 1.5 = 4; imaginary: −1 − 3 + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

Find the complex number w for which
(4 − 7i) + w = −2 + 3i.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

46

QUESTION ID
S2-M-Q5

TYPE
Short answer

CORRECT ANSWER
−6 + 10i

GRADING NOTE Form required: a + bi. w = (−2 + 3i) − (4 − 7i) = −6 + 10i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify (3 + 5x) + (−7 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

48

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−4 + 7x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

Simplify (6 − 4x) − (−2 + 9x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

49

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8 − 13x

GRADING NOTE Any equivalent form. The minus sign applies to both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

(2 − 3y) − (5 − y) equals

A. 7 − 4y

B. −3 + 2y

C. −3 − 4y

D. −3 − 2y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

50

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3 − 2y

GRADING NOTE 2 − 5 = −3 and −3y + y = −2y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$(4 + 2t) + (1 - 6t) = 5 - 4t.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

51

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4 + 1 = 5$ and $2t - 6t = -4t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (0.5 − x) − (1.5 + 2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

52

QUESTION ID

S2-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−1 − 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 2

Add and subtract integers and decimals • Grade 7 • The Number System • 7.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $6 - (-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

54

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-4 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

55

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

$-1 - 3 + 4$ equals

A. 8

B. 0

C. -8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

56

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE $-1 - 3 = -4$ and $-4 + 4 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 - (-1.5) = 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

57

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2.5 + 1.5 = 4.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

58

QUESTION ID

S2-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 3

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−16) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

60

QUESTION ID

S2-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

Write √(−9) in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

61

QUESTION ID

S2-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-B3-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

√(−25) equals

A. −5

B. 25i

C. 5i

D. −5i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

62

QUESTION ID

S2-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5i

GRADING NOTE √(−25) = √25 · i = 5i (the principal root).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-B3-Q3**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{(-1)} = i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

63

QUESTION ID

S2-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of i as the principal square root of −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $-\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

64

QUESTION ID

S2-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 1

Find real parameters that make a sum real or imaginary • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

k is a real number. For what value of k is (3 + ki) + (k − 5i) a real number? Give that real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

66

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

k = 5; the sum is 8

GRADING NOTE Form required: k and the number. The imaginary part k − 5 must be 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

k is real. For what value of k is
(2k − 6) + 4i − (k + 1 − i) a pure imaginary number
(real part 0)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

67

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. Real part: 2k − 6 − k − 1 = k − 7 = 0. The sum is then 5i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

True or false?

The sum of a complex number and its opposite is always 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

68

QUESTION ID

S2-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (a + bi) + (−a − bi) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

a is real and (a + 3i) − (2 + ai) is real. What is a?

A. 0

B. 3

C. 2

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

69

QUESTION ID

S2-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

GRADING NOTE Imaginary part 3 − a = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

Explain why the sum of a + bi and a − bi is always a real number when a and b are real.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

70

QUESTION ID

S2-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The imaginary parts b and −b add to 0, leaving 2a, a real number.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 2

Add complex numbers as vectors in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5

Plot 2 + i and 1 + 3i as points in the complex plane (real axis horizontal). What point represents their sum?

Sum point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

72

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4), the number 3 + 4i

GRADING NOTE Form required: the point or the number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

The points 0, 2 + i, 1 + 3i and their sum are vertices of a quadrilateral. What kind of quadrilateral is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

73

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A parallelogram

GRADING NOTE Any equivalent answer. Adding complex numbers adds their vectors (parallelogram rule).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

True or false?

Subtracting w from z gives the vector from the point w to the point z.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

74

QUESTION ID

S2-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE z − w is the displacement from w to z.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

What is the distance between the points 5 + 2i and 1 − i in the complex plane?

A. 25

B. 5

C. √13

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

75

QUESTION ID

S2-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE |(5 + 2i) − (1 − i)| = |4 + 3i| = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

Find the midpoint of the segment joining $6 - 2i$ and $-2 + 4i$, as a complex number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

76

QUESTION ID

S2-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$2 + i$

GRADING NOTE Form required: a + bi. The average $((6 - 2i) + (-2 + 4i))/2 = 2 + i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

Multiply complex numbers and simplify powers of i

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Multiply binomials	**HSA-APR.A.1**
BACKWARD 2	Use $i^2 = -1$ and the cycle of powers of i	**HSN-CN.A.1**
BACKWARD 3	Divide whole numbers to find a remainder	**4.NBT.B.6**
FORWARD 1	Find a real parameter that makes a product real	**HSN-CN.A.2**
FORWARD 2	Divide complex numbers using conjugates	**HSN-CN.A.3**

SET 3 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply (3 + 2i)(4 − 5i). Write the product in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

79

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

22 − 7i

GRADING NOTE Form required: a + bi. $12 - 15i + 8i - 10i^2 = 12 - 7i + 10 = 22 - 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

Simplify $(-2 + i)^2$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

80

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

3 − 4i

GRADING NOTE Form required: a + bi. $4 - 4i + i^2 = 4 - 4i - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

Multiply $(5 - 3i)(5 + 3i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

81

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

34 (that is, 34 + 0i)

GRADING NOTE Form required: $a + bi$ or the real number 34. $25 + 15i - 15i - 9i^2 = 25 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

Simplify $(1 + i)^3$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

82

QUESTION ID

S3-M-Q4

TYPE

Short answer

CORRECT ANSWER

−2 + 2i

GRADING NOTE Form required: a + bi. $(1 + i)^2 = 2i$, and $2i(1 + i) = 2i + 2i^2 = -2 + 2i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

Simplify $i^{23} \cdot (4 - i)$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

83

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-1 - 4i$

GRADING NOTE Form required: a + bi. $23 = 4 \cdot 5 + 3$, so $i^{23} = i^3 = -i$; $-i(4 - i) = -4i + i^2 = -1 - 4i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 1

Multiply binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply (3 + 2x)(4 − 5x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

85

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$12 - 7x - 10x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(-2 + x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

86

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$4 - 4x + x^2$

GRADING NOTE Any equivalent form. Not $x^2 + 4$ (missing the middle term).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

(5 − 3x)(5 + 3x) equals

A. $25 - 9x^2$

B. $25 + 9x^2$

C. $10 - 9x^2$

D. $25 - 30x - 9x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

87

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $25 - 9x^2$

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$(1 + x)^2 = 1 + x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

88

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(1 + x)^2 = 1 + 2x + x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

Multiply x(4 − x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

89

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$4x - x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B1-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 2

Use $i^2 = -1$ and the cycle of powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

91

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. Definition of i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify i^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

92

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-i$

GRADING NOTE Any equivalent form. $i^3 = i^2 \cdot i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

Simplify i^4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

93

QUESTION ID

S3-B2-Q3

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. $i^4 = (i^2)^2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-10i^2 = -10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

94

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^2 = -1$, so $-10i^2 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

Which equals 1?

A. i^9

B. i^8

C. i^3

D. i^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

95

QUESTION ID

S3-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. i^8

GRADING NOTE Powers of i repeat every 4; 8 is a multiple of 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 3

Divide whole numbers to find a remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the remainder when 23 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

97

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 23 = 4 · 5 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

Find the remainder when 50 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

98

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 50 = 4 · 12 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

What is the remainder when 36 is divided by 4?

A. 0

B. 4

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

99

QUESTION ID

S3-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0

GRADING NOTE $36 = 4 \cdot 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

True or false?

The remainder when 101 is divided by 4 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

100

QUESTION ID

S3-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $101 = 4 \cdot 25 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

Write 39 as 4q + r with 0 ≤ r < 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

101

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

39 = 4 · 9 + 3

GRADING NOTE Form required: q = 9 and r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 1

Find a real parameter that makes a product real • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

a is a real number. For what value of a is (a + 2i)(3 − i) a real number? Give the product.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

103

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

a = 6; the product is 20

GRADING NOTE Form required: both. (a + 2i)(3 − i) = (3a + 2) + (6 − a)i; the imaginary part 6 − a = 0. Product 18 + 2 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

b is real. For what value of b is (2 + bi)(1 + i) pure imaginary (real part 0)?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

104

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Product: (2 − b) + (2 + b)i; real part 2 − b = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

True or false?

The product of two non-real complex numbers can be a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

105

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example (1 + i)(1 − i) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

For real c, $(c + i)^2$ is real when

A. c = 1

B. c = 0

C. c = −1

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

106

QUESTION ID

S3-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. c = 0

GRADING NOTE $(c + i)^2 = c^2 − 1 + 2ci$; imaginary part $2c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

Show that (a + bi)(a − bi) is a real number for all real a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

107

QUESTION ID

S3-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$a^2 - abi + abi - b^2i^2 = a^2 + b^2$, which is real.

GRADING NOTE Any equivalent algebra.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 2

Divide complex numbers using conjugates • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write (3 + 4i) / (1 + 2i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

109

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

11/5 − (2/5)i

GRADING NOTE Form required: a + bi. Multiply by (1 − 2i)/(1 − 2i): (3 + 4i)(1 − 2i)/5 = (11 − 2i)/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

Write 10 / (3 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

110

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 + i

GRADING NOTE Form required: a + bi. 10(3 + i)/10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

True or false?

To divide by 2 − 5i, multiply numerator and denominator by 2 + 5i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

111

QUESTION ID

S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The conjugate makes the denominator real: 4 + 25 = 29.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

(2 + i) / i equals

A. −1 + 2i

B. 1 − 2i

C. 2 + i

D. 1 + 2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

112

QUESTION ID

S3-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1 − 2i

GRADING NOTE (2 + i)(−i)/(i · −i) = (−2i + 1)/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

Explain why multiplying by the conjugate of the denominator always gives a real denominator.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

113

QUESTION ID

S3-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$(c + di)(c - di) = c^2 + d^2$, a real number (and positive unless $c = d = 0$).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSN-CN.C.7 • THE COMPLEX NUMBER SYSTEM

Solve quadratic equations with complex solutions

MAIN	Algebra II main question	**HSN-CN.C.7**
BACKWARD 1	Use the quadratic formula and the discriminant	**HSA-REI.B.4.b**
BACKWARD 2	Write square roots of negative numbers with i	**HSN-CN.A.1**
BACKWARD 3	Solve equations of the form $(x - h)^2 = k$	**HSA-REI.B.4.b**
FORWARD 1	Find a quadratic from one complex solution	**HSN-CN.C.7**
FORWARD 2	Classify solutions for a family of quadratics	**HSN-CN.C.7**

SET 4 • HSN-CN.C.7

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 49 = 0$ over the complex numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

116

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7i or x = −7i

GRADING NOTE Form required: both solutions, exact. $x^2 = -49$, $x = \pm 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

Solve $x^2 - 4x + 13 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

117

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2 + 3i or

x = 2 − 3i

GRADING NOTE Form required: both, a + bi. Discriminant 16 − 52 = −36; x = (4 ± 6i)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

Solve $2x^2 + 2x + 5 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

118

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = −1/2 + (3/2)i or
x = −1/2 − (3/2)i

GRADING NOTE Form required: both, a + bi. Discriminant 4 − 40 = −36; x = (−2 ± 6i)/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

Solve $(x + 3)^2 = -20$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

119

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

$x = -3 + 2\sqrt{5}\,i$ or
$x = -3 - 2\sqrt{5}\,i$

GRADING NOTE Form required: both, exact. $x + 3 = \pm\sqrt{-20} = \pm 2\sqrt{5}\,i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

Solve $3x^2 - 6x + 7 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

120

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 1 + (2\sqrt{3}/3)i$ or $x = 1 - (2\sqrt{3}/3)i$

GRADING NOTE Form required: both, exact, a + bi. Discriminant $36 - 84 = -48 = (4\sqrt{3}\, i)^2$; $x = (6 \pm 4\sqrt{3}\, i)/6$. Check: product 7/3 and sum 2 agree with c/a and −b/a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 1

Use the quadratic formula and the discriminant • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Find the discriminant $b^2 - 4ac$ of $x^2 - 4x + 13 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

122

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 16 − 52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B1-Q1**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the discriminant of $2x^2 + 2x + 5 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

123

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−36

GRADING NOTE Any equivalent form. 4 − 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

True or false?

If the discriminant is negative, the quadratic equation has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

124

QUESTION ID

S4-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √ of a negative number is not real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In $3x^2 - 6x + 7 = 0$, the values of a, b and c are

A. 3, −6, −7

B. 3, 6, 7

C. −3, −6, 7

D. 3, −6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

125

QUESTION ID

S4-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3, −6, 7

GRADING NOTE Standard form $ax^2 + bx + c = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

Write the quadratic formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

126

QUESTION ID

S4-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 2

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write $\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

128

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6i

GRADING NOTE Form required: bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write √(−48) in simplest form with i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

129

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4√3 i

GRADING NOTE Form required: simplest radical form. 48 = 16 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

√(−20) equals

A. 2√5 i

B. 4√5 i

C. −2√5

D. 10i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

130

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2√5 i

GRADING NOTE √20 = 2√5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{(-49)} = -7.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

131

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(-49)} = 7i.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

Write (−2 + √(−36)) / 4 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

132

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2 + (3/2)i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B2-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 3

Solve equations of the form $(x - h)^2 = k$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $(x - 2)^2 = 9$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

134

QUESTION ID

S4-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = 5 or x = −1

GRADING NOTE Form required: both. $x - 2 = \pm 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-B3-Q1**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

Solve $(x + 1)^2 = 5$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

135

QUESTION ID

S4-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x = -1 + \sqrt{5}$ or $x = -1 - \sqrt{5}$

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving $x^2 = 16$ gives only $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

136

QUESTION ID

S4-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Also $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

The solutions of $(x + 4)^2 = 25$ are

A. 5 and −5

B. −1 and 9

C. 1 and −9

D. 21 and −29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

137

QUESTION ID

S4-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1 and −9

GRADING NOTE $x + 4 = \pm 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

Solve $x^2 = 50$ over the real numbers.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

138

QUESTION ID

S4-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 5\sqrt{2}$ or $x = -5\sqrt{2}$

GRADING NOTE Form required: both, exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 1

Find a quadratic from one complex solution • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

$3 - 2i$ is a solution of $x^2 + bx + c = 0$, where b and c are real. Find b and c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

140

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

b = −6, c = 13

GRADING NOTE Form required: both. With real coefficients the other solution is 3 + 2i; sum 6 = −b, product 9 + 4 = 13 = c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

$-1 + 4i$ is a solution of a quadratic equation $x^2 + bx + c = 0$ with real b and c. Write the equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

141

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x + 17 = 0$

GRADING NOTE Any equivalent equation. Sum −2, product 1 + 16 = 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

True or false?

If 2 + i is a solution of a quadratic with real coefficients, then 2 − i is also a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

142

QUESTION ID

S4-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Non-real solutions of real quadratics come in conjugate pairs (± in the formula).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

Which quadratic with real coefficients has 5i as a solution?

A. $x^2 + 25 = 0$

B. $x^2 + 5 = 0$

C. $x^2 - 25 = 0$

D. $x^2 - 10x + 25 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

143

QUESTION ID

S4-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 25 = 0$

GRADING NOTE $(5i)^2 + 25 = -25 + 25 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

Explain why $x^2 - (2 + i)x = 0$ has the solution $2 + i$ but not the solution $2 - i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

144

QUESTION ID

S4-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Its coefficient 2 + i is not real, so solutions need not come in conjugate pairs: the solutions are 0 and 2 + i.

GRADING NOTE Any equivalent explanation naming the non-real coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 2

Classify solutions for a family of quadratics • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

For which real values of k does $x^2 + kx + 9 = 0$ have two non-real solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

146

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-6 < k < 6$

GRADING NOTE Form required: an inequality. Discriminant $k^2 - 36 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

For which real values of c does $x^2 - 4x + c = 0$ have non-real solutions? Give them for c = 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

147

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

c > 4; for c = 8, x = 2 ± 2i

GRADING NOTE Form required: both parts. Discriminant $16 - 4c < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

True or false?

$x^2 + kx - 1 = 0$ has non-real solutions for some real k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

148

QUESTION ID

S4-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its discriminant $k^2 + 4$ is always positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

$x^2 + 6x + m = 0$ has exactly one (repeated) real solution when m =

A. 9

B. 3

C. 36

D. −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

149

QUESTION ID

S4-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 9

GRADING NOTE $36 - 4m = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

For c > 4, write the solutions of $x^2 - 4x + c = 0$ in terms of c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

150

QUESTION ID

S4-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$x = 2 \pm i\sqrt{(c - 4)}$

GRADING NOTE Form required: exact in terms of c. $(4 \pm \sqrt{(16 - 4c)})/2 = 2 \pm \sqrt{(4 - c)} = 2 \pm i\sqrt{(c - 4)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

Factor polynomials over the complex numbers

MAIN	Algebra II main question	**HSN-CN.C.8**
BACKWARD 1	Factor a difference of squares	**HSA-SSE.A.2**
BACKWARD 2	Multiply complex conjugates	**HSN-CN.A.2**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Factor cubic and quartic polynomials over the complex numbers	**HSN-CN.C.8**
FORWARD 2	Write polynomials with real coefficients from complex zeros	**HSN-CN.C.9**

SET 5 • HSN-CN.C.8

MAIN

Algebra II main question • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

153

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 5i)(x - 5i)$

GRADING NOTE Any order of factors. $(x + 5i)(x - 5i) = x^2 - 25i^2 = x^2 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q1**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

Factor $x^2 + 7$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

154

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + i\sqrt{7})(x - i\sqrt{7})$

GRADING NOTE Any order. $x^2 - (i\sqrt{7})^2 = x^2 + 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q2**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

Factor $4x^2 + 9$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

155

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(2x + 3i)(2x - 3i)$

GRADING NOTE Any equivalent factorization (for example $4(x + 3i/2)(x - 3i/2)$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q3**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

Factor $x^4 - 16$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

156

QUESTION ID

S5-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 2)(x - 2i)(x + 2i)$

GRADING NOTE Any order. $x^4 - 16 = (x^2 - 4)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q4**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

Factor $x^2 - 6x + 13$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

157

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - (3 + 2i))(x - (3 - 2i))$

GRADING NOTE Any equivalent form. Its zeros are $3 \pm 2i$ (discriminant -16). Check: sum 6, product $9 + 4 = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-M-Q5**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 1

Factor a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 49$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

159

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 7)(x − 7)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

Factor $9x^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

160

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(3x + 2)(3x − 2)

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

$x^4 - 1$ equals

A. $(x - 1)^4$

B. $(x^2 - 1)^2$

C. $(x^2 + 1)^2$

D. $(x^2 - 1)(x^2 + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

161

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $(x^2 - 1)(x^2 + 1)$

GRADING NOTE $(x^2)^2 - 1^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

162

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

Factor $x^2 - 5$ using radicals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

163

QUESTION ID

S5-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + \sqrt{5})(x - \sqrt{5})$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 2

Multiply complex conjugates • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(x + 5i)(x - 5i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

165

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 25$

GRADING NOTE Any equivalent form. $-25i^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

Multiply $(2x + 3i)(2x - 3i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

166

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$4x^2 + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

True or false?

$(3i)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

167

QUESTION ID

S5-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $9i^2 = -9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - (1 + i))(x - (1 - i))$ equals

A. $x^2 + 2x + 2$

B. $x^2 + 2$

C. $x^2 - 2x$

D. $x^2 - 2x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

168

QUESTION ID

S5-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 - 2x + 2$

GRADING NOTE Sum 2, product 1 + 1 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

Simplify $(i\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

169

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B2-Q5**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 25 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

171

QUESTION ID

S5-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±5i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 6x + 13 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

172

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 ± 2i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-B3-Q2**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

True or false?

If r is a zero of p(x), then x − r is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

173

QUESTION ID

S5-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Factor theorem, also for complex r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

The zeros of $x^2 + 4$ are

A. 4i and −4i

B. −4

C. 2i and −2i

D. 2 and −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

174

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2i and −2i

GRADING NOTE $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x + 5 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

175

QUESTION ID

S5-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x = −1 ± 2i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 1

Factor cubic and quartic polynomials over the complex numbers • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^3 + 8$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

177

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 2)(x - 1 - i\sqrt{3})(x - 1 + i\sqrt{3})$

GRADING NOTE Any order. $x^3 + 8 = (x + 2)(x^2 - 2x + 4)$, and $x^2 - 2x + 4$ has zeros $1 \pm i\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

Factor $x^3 + 9x$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

178

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x(x + 3i)(x - 3i)$

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q2**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

Factor $x^4 + 5x^2 + 4$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

179

QUESTION ID

S5-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$(x + i)(x - i)(x + 2i)(x - 2i)$

GRADING NOTE Any order. $(x^2 + 1)(x^2 + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q3**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

True or false?

$x^3 - 1 = (x - 1)(x^2 + x + 1)$, and $x^2 + x + 1$ has two non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

180

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Discriminant $1 - 4 = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

Which is a linear factor of $x^3 - 27$ over the complex numbers?

A. $x + 3i$

B. $x + 3$

C. $x - (-3/2 + (3\sqrt{3}/2)i)$

D. $x - 3i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

181

QUESTION ID

S5-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C.

$x - (-3/2 + (3\sqrt{3}/2)i)$

GRADING NOTE $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$; $x^2 + 3x + 9$ has zeros $-3/2 \pm (3\sqrt{3}/2)i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 2

Write polynomials with real coefficients from complex zeros • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9

Write a polynomial with real coefficients, leading coefficient 1 and least degree, whose zeros include 2i and 1 − i.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

183

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$$(x^2 + 4)(x^2 - 2x + 2) = x^4 - 2x^3 + 6x^2 - 8x + 8$$

GRADING NOTE Any equivalent form. Conjugates −2i and 1 + i must also be zeros, so degree 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

What is the least degree of a polynomial with real coefficients that has zeros 3, 4i and 2 + i?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

184

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. −4i and 2 − i are also zeros: 1 + 2 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q2**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

True or false?

A polynomial with real coefficients can have exactly one non-real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

185

QUESTION ID

S5-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Non-real zeros come in conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

A cubic with real coefficients has zeros 5 and −i. Its third zero is

A. −5

B. 5i

C. i

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

186

QUESTION ID

S5-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. i

GRADING NOTE Conjugate of −i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

Explain why a polynomial with real coefficients and degree 3 must have at least one real zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

187

QUESTION ID

S5-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It has 3 complex zeros with multiplicity; non-real zeros pair up, so they come two at a time, leaving at least one real zero.

GRADING NOTE Any equivalent explanation using the Fundamental Theorem of Algebra and conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

Use the Fundamental Theorem of Algebra

MAIN	Algebra II main question	**HSN-CN.C.9**
BACKWARD 1	Find the degree of a polynomial	**HSA-SSE.A.1.a**
BACKWARD 2	Find zeros from factored form	**HSA-SSE.B.3.a**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Relate the complex zeros to the real x-intercepts of a graph	**HSA-APR.B.3**
FORWARD 2	Prove existence statements about real zeros	**HSN-CN.C.9**

SET 6 • HSN-CN.C.9

MAIN

Algebra II main question • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does $p(x) = 3x^5 - x^2 + 7$ have? Name the theorem you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

190

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

5, by the Fundamental Theorem of Algebra (degree 5)

GRADING NOTE Form required: the number and the theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

List every complex zero of $p(x) = (x - 2)^3(x^2 + 9)$ with its multiplicity, and check that the multiplicities add to the degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

191

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

2 (multiplicity 3), 3i (multiplicity 1), −3i (multiplicity 1);
3 + 1 + 1 = 5
= degree

GRADING NOTE Form required: all zeros with multiplicities and the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

Show that the Fundamental Theorem of Algebra holds for every quadratic $p(x) = ax^2 + bx + c$ with real a, b, c and $a \neq 0$: p has exactly two complex zeros counted with multiplicity.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

192

QUESTION ID

S6-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The zeros are $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$: two real zeros if $D > 0$, one real zero of multiplicity 2 if $D = 0$, two conjugate non-real zeros if $D < 0$.

GRADING NOTE REQUIRED WORK: Use the quadratic formula and the cases of the discriminant.

Any valid proof or derivation. Essential: the formula gives all zeros; the three cases of $D = b^2 - 4ac$; for $D < 0$, $\sqrt{D} = i\sqrt{(-D)}$ is a complex number; for $D = 0$ the zero $-b/(2a)$ has multiplicity 2 since $p(x) = a(x + b/(2a))^2$. Invalid: checking one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q3**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

Find every complex zero of $p(x) = x^4 - 81$ and check that the count agrees with the Fundamental Theorem of Algebra.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

193

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

3, −3, 3i, −3i; four zeros for degree 4

GRADING NOTE Form required: all four and the check. $(x^2 - 9)(x^2 + 9)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

A student says $p(x) = x^3 + x + 1$ has no zeros because it does not factor with integers. Use the Fundamental Theorem of Algebra to explain how many complex zeros it has, and explain why at least one of them is real.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

194

QUESTION ID

S6-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

It has 3 complex zeros with multiplicity; non-real zeros of a real polynomial come in conjugate pairs, so at most 2 are non-real and at least 1 is real.

GRADING NOTE REQUIRED WORK: Count zeros by degree; use conjugate pairs for real coefficients.

Any valid explanation. Essential: degree 3 gives 3 zeros; "no integer factorization" does not mean "no zeros"; conjugate pairs force an odd number of real zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q5**

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 1

Find the degree of a polynomial • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

What is the degree of $3x^5 - x^2 + 7$?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

196

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

What is the degree of $(x - 2)^3(x^2 + 9)$ when multiplied out?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

197

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 3 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

The degree of $x(x + 1)(x^2 - 4)$ is

A. 1

B. 2

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

198

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

GRADING NOTE 1 + 1 + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

True or false?

The degree of $4x^3 + 2x^4$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

199

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The highest power is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the leading coefficient of $-2x^3 + 5x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

200

QUESTION ID

S6-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 2

Find zeros from factored form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the zeros of $(x - 2)(x + 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

202

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 and −5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

Find the zeros of $(x - 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

203

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1 (repeated three times)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

True or false?

A product is 0 only when one of its factors is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

204

QUESTION ID

S6-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Zero-product property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

The zeros of x(x + 4) are

A. 0 and 4

B. 0 and −4

C. −4 only

D. 4 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

205

QUESTION ID

S6-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0 and −4

GRADING NOTE x = 0 or x + 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the zeros of $(2x - 3)(x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

206

QUESTION ID

S6-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3/2 and −1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 9 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

208

QUESTION ID

S6-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x = ±3i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-B3-Q1**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x + 2 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

209

QUESTION ID

S6-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = −1 ± i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 + 1 = 0$ has no complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

210

QUESTION ID
S6-B3-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It has ±i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

The solutions of $x^2 - 2x + 5 = 0$ are

A. $1 \pm 4i$

B. $-1 \pm 2i$

C. $2 \pm i$

D. $1 \pm 2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

211

QUESTION ID

S6-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1 ± 2i

GRADING NOTE Discriminant −16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = -81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

212

QUESTION ID

S6-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x = ±9i

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 1

Relate the complex zeros to the real x-intercepts of a graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A polynomial of degree 4 with real coefficients has a graph that crosses the x-axis at exactly two points and touches it nowhere else. Each crossing is a simple zero. How many non-real zeros does it have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

214

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 4 zeros in all, 2 real (the crossings), so 2 non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

The graph of $y = x^3 - 4x^2 + 5x - 2$ touches the x-axis at $x = 1$ and crosses it at $x = 2$. List all complex zeros with multiplicity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

215

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1 (multiplicity 2) and 2 (multiplicity 1)

GRADING NOTE Form required: zeros with multiplicities. $(x - 1)^2(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 6 whose graph never meets the x-axis has 6 non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

216

QUESTION ID

S6-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All 6 zeros are non-real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

A degree-5 polynomial with real coefficients has exactly one real zero, of multiplicity 1. How many non-real zeros does it have?

A. 5

B. 2

C. 4

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

217

QUESTION ID

S6-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE $5 - 1 = 4$ (two conjugate pairs).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

Can a degree-4 polynomial with real coefficients have exactly three real zeros, each of multiplicity 1? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

218

QUESTION ID

S6-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No: the fourth zero would be a single non-real zero, but non-real zeros come in conjugate pairs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 2

Prove existence statements about real zeros • High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9

Prove that every polynomial of odd degree with real coefficients has at least one real zero.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

220

QUESTION ID

S6-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

An odd number of zeros, minus an even number of non-real zeros, leaves an odd (so nonzero) number of real zeros.

GRADING NOTE REQUIRED WORK: Count zeros with the theorem; pair non-real zeros.

Any valid proof or derivation. Essential: n zeros with multiplicity (n odd); conjugate pairs have equal multiplicity, so the non-real zeros count is even; odd minus even is odd ≥ 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

A degree-4 polynomial with real coefficients has the zeros 1 + i and 3. What are all of its possible zero lists?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

221

QUESTION ID
S6-F2-Q2

TYPE
Short answer

CORRECT ANSWER
1 + i, 1 − i, 3 and one more real zero r (r may be 3 again, giving 3 with multiplicity 2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

True or false?

A degree-4 polynomial with real coefficients may have no real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

222

QUESTION ID

S6-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example $x^4 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

A polynomial with real coefficients has degree 7. The possible numbers of real zeros (with multiplicity) are

A. 7 only

B. 0, 2, 4, 6

C. 1, 3, 5, 7

D. 0 to 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

223

QUESTION ID
S6-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 1, 3, 5, 7

GRADING NOTE The number of real zeros is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

Explain why the Fundamental Theorem of Algebra says nothing about whether a polynomial's zeros can be found exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

224

QUESTION ID
S6-F2-Q5

TYPE
Short answer

CORRECT ANSWER
It guarantees that the zeros exist and counts them; it gives no method or formula for finding them.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Algebra II ((+) standard listed in the pathway) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5